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IT FDN 110 GP

Assignment 5

Assignment 05

Introduction

This week we expanded upon Assignment 4, with the primary addition being the use of dictionaries and exception handling. The new version stores each student as a dictionary instead of a list, saves and loads its data in a JSON file using Python's json module, and uses structured error handling to deal with bad input and file problems. I also posted my files to a public GitHub repository so other students can review them. These can be found at (https://github.com/crphelm93/Intro-To-Python) .

Creating The Script

I started by updating the header and importing json. I then changed the file from enrollments.csv to enrollments.json. I also changed the variables, such as student_data. I changed this from a list to a dictionary, which allowed for looking up values by name. I also changed menu_choice to an empty string.

```
import json

# Define the Data Constants
MENU: str = '''
---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course
    2. Show current data
    3. Save data to a file
    4. Exit the program
-----------------------------------------
'''
FILE_NAME: str = "Enrollments.json"
```

The starter read the CSV file line by line, splitting each line on commas. With JSON, that whole loop became one call. json.load() reads the file and turns it straight into a list of dictionaries. Before testing, I added two starting records to enrollments.json to fix the error I was getting.

```
try:
    file = open(FILE_NAME, "r")
    students = json.load(file)
```

I then tested registering a student. This was similar to each of the last 3 assignments, with first name, last name, and course name. I then needed to ask the program to store the data. Since the assignment asked for both a readable message and comma separated values (a mistake I made on Assignment 4), I loop through the table twice, using keys instead of index numbers to pull out each value.

```
for student in students:
    print(f'Student {student["FirstName"]} {student["LastName"]} is enrolled in {student["CourseName"]}')
print("-" * 50)
for student in students:
    print(f'{student["FirstName"]},{student["LastName"]},{student["CourseName"]}')
```

I hit a bit of a wall with reading the file, and needed some help from Claude in explaining and building out the try, FileNotFoundError, and Exception block. (Anthropic. (2026). *Claude Opus 5.5* (medium effort; model claude-opus-5-5) [Large language model]. https://claude.ai). Claude suggested other changes as well, such as the isalpha() method as the student names do not include mixed characters and are strictly numbers.

```
# Save the data to a file
elif menu_choice == "3":
    try:
        file = open(FILE_NAME, "w")
        json.dump(students, file, indent=2)
        file.close()
        print("The following data was saved to file!")
        for student in students:
            print(f'Student {student["FirstName"]} {student["LastName"]} is enrolled in {student["CourseName"]}')
    except TypeError as e:
        print("Please check that the data is a valid JSON format\n")
        print("-- Technical Error Message -- ")
        print(e, e.__doc__, type(e), sep='\n')
    except Exception as e:
        print("-- Technical Error Message -- ")
        print("Built-In Python error info: ")
        print(e, e.__doc__, type(e), sep='\n')
    finally:
        if file is not None and file.closed == False:
            file.close()
    continue
```

I then ran my tests. Option one and two worked, just as they had in weeks prior. After choosing option 3, I opened enrollments.json in PyCharm to confirm that the new students were saved correctly. I then restarted the program and chose option 2 to make sure the saved data loaded back in. I did not want to make the same mistake as last week so I tested the script in Claude as

well, who confirmed that I had met all the acceptance criteria.

> The program itself meets every code criterion. A few items on the list depend on things only you can do: running it in PyCharm, the GitHub repository and posting the link.

Summary/Conclusion

I am a bit unsure about my code in comparison to other's, as Heidi's was much simpler regarding saving the data to a file. This is the first assignment where I did feel like I was truly struggling, and while I enjoy challenges I am uncertain. She used while True: under the user data input and I did not, and I am starting to second guess my code. Perhaps it was the busy week or feeling rushed. Regardless, I was able to get the script to function so perhaps this is a stylistic difference? I have never shared my own files on Github either, so the uploading and organizational process was a bit tricky at first. As someone new to sharing code and interacting with others in a collaborative setting, this is certainly fun and intriguing and I look forward to what the coming weeks have in store.

References

Anthropic. (2026). *Claude Opus 5.5* (medium effort; model claude-opus-5-5) [Large language model]. https://claude.ai

“Mod05 - ExceptionHandling.” YouTube, uploaded by Anubhaw Arya, https://www.youtube.com/watch?v=O46YoSo477Y

“Mod05 - UsingDictionariesAndFiles.” YouTube, uploaded by Anubhaw Arya, https://www.youtube.com/watch?v=Kgh2TZ0tf28

“Mod05 -UsingJsonFiles.” YouTube, uploaded by Anubhaw Arya, https://www.youtube.com/watch?v=XDDyhzRFDuc